

10 (a) Motor neurones and sensory neurones have different roles in nervous coordination.

Contrast the structure **and** function of sensory neurones and motor neurones.

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(b) Mammals have many types of neurones, which vary in axon diameter and myelination.

Table 10.1 shows the axon diameter and mean impulse transmission speed of four different types of mammalian neurone.

Table 10.1

neurone type	myelination	mean axon diameter / μm	mean impulse transmission speed/ m s^{-1}
motor α	✓	13.0–20.0	80–120
motor β	✓	6.0–12.0	35–75
motor δ	✓	1.0–5.0	5–35
motor C	x	0.2–1.5	0.5–2.0

Using the data shown in Table 10.1, comment on the relationship between:

- myelination and mean impulse transmission speed
- axon diameter and mean impulse transmission speed.

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(c) Fig. 10.1 is a graph of an action potential in a mammalian neurone.

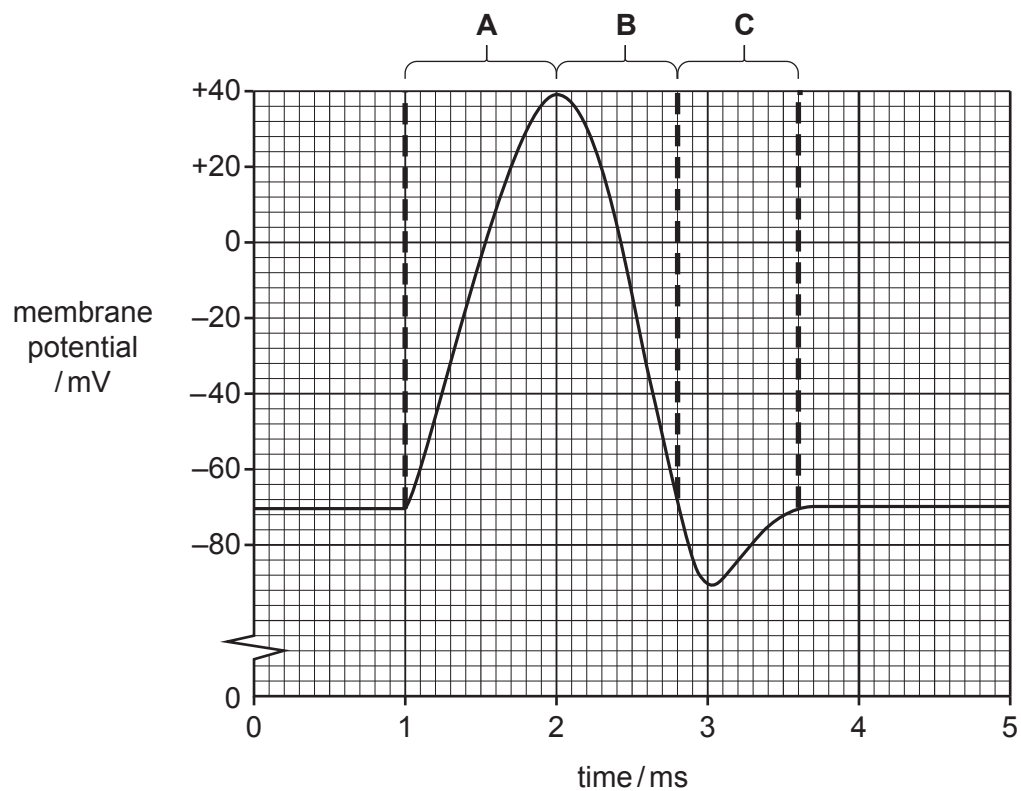


Fig. 10.1

With reference to Fig. 10.1 suggest why:

- no further action potential can occur during **A** and **B**
- it is difficult for a further action potential to occur during **C**.

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